# PA194Q5 and PA194Q6

A comparison of the PA consistency-machine corpus
and two 194-state constructions

Construction and comparison report

13 November 2026, Pacific time

**Abstract**

The state-count target below 194 was not achieved. The delivered fallback consists of two explicit, complete, binary, one-tape, blank-start machines, PA194Q5 and PA194Q6, each with 194 working states and 388 transition rows. They combine the exchanged quantifier-distribution head with formula coefficients five and six, respectively, while retaining the original sum-testing Shift and dummy-push modus-ponens controller. A simultaneous eight-fold quotient has budget $135 + 67 - 8 = 194$. Both tables pass complete source-step contracts and exact whole-tape initialization. A separate literal-tape implementation and a finite implicational proof certificate are supplied.

The comparison is pinned to ProveIt commit `317b07268681b44ff352d173a162e2a8c241b2fe`. It covers 51 placed research packets and three incoming archives, including explicit JSON machine encodings. Forty existing 194-state graph classes were identified. Neither new table is equivalent to any of them under state renaming, choice of spanning root, tape reflection, symbol complementation, or disregard of the final write and move. The ordinary mathematical argument establishes the PA-provable consistency equivalence using the attributed radix-calibration and source-compiler lemmas. No new end-to-end Lean, Rocq, or iPRA equivalence theorem is claimed.

## 1 Result and exact scope

The requested improvement would have been a verified machine with at most 193 working states. None was obtained. The positive result is the following pair of new constructions, using the permitted 194-state fallback.

**Theorem 1** (Ordinary mathematical calibration)**.** *For $k \in \{5, 6\}$, let $M_k$ be the literal table `machines/PA194Qk/machine.tm`, with the letter `k` replaced by the indicated digit. Then*

$$\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(M_k, \mathit{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})).$$

*The proof uses the inherited ordinary PA calibration stated in Section 3, the explicit source/compiler transport below, and the delivered finite and symbolic physical certificates.*

Here Con(PA) is the usual arithmetical assertion that there is no finite PA proof of $0 = 1$ under a fixed effective proof coding. The theorem is a statement about what PA proves, not an assumption that PA is sound or consistent. Its proof does not use a PA reflection principle.

Each machine starts in state 0 with its head at cell 0 on a two-way infinite all-zero tape. The alphabet is $\{0, 1\}$. Working states are $0, \ldots, 193$; 194 is the separate halt label and has no rows. Every transition writes one bit and moves exactly one cell left or right. There are no stationary moves, preloaded input, auxiliary tapes, or uncounted working states. All 194 working states are reachable in the transition graph; this last fact is a graph property, not a claim that each is visited by the particular blank computation.

| Quantity | PA194Q5 | PA194Q6 |
|---|---|---|
| Working states / transition rows | 194 / 388 | 194 / 388 |
| Unfolded controls | 202 | 202 |
| Fixed controls / AX readers | 135 / 67 | 135 / 67 |
| Simultaneous folds / local cases | 8 / 32 | 8 / 32 |
| Primitive source sites | 3,734 | 3,795 |
| Primitive branch contracts | 4,841 | 4,902 |
| Maximum source-address length | 35 | 35 |
| Literal blank-startup steps | 58,320 | 58,320 |

The full dense tables appear in Appendix C. Their SHA-256 digests are:

**PA194Q5**

`150041f819135fa32b4521154823f1b96e635e8141329a6e8fbf15988baa073a`

**PA194Q6**

`7f53cacaa0533e601bad9685957a1ef5fede892533e0d62bd49684ef7cf984d4`

The underlying ideas of premise exchange and coefficient recoding already occur separately in the supplied corpus. This report claims their new literal combinations, their complete source and physical binding, and the resulting calibration. It makes no claim to invent either component or to set a new state-count record.

## 2 What was compared

The four requested repository areas were inspected at the pinned commit: PA research, incoming research packets, ZFC research, and ConciseIPRA. The PA area contains 51 placed packets; three additional incoming ZIPs are `PA194NS_research.zip`, `pa194qn_research_2026-11-13.zip`, and `pa194haz_research_2026-11-14.zip`. Their filenames' dates are preserved even where they use UTC rather than the date on this report.

The repository's own historical manifest lists 4,984 delivered `.tm` occurrences before deduplication. This research inventories the *current* placed files and the nested incoming archives, so the following counts use a different population and classification rule. The historical and current occurrence counts should not be subtracted.

| Current audit quantity | Count |
|---|---|
| Principal research packets | 54 |
| Recognized literal one-halt binary table occurrences | 1,402 |
| Byte identities among those occurrences | 721 |
| Spanning-root action-graph classes among literal tables | 497 |
| JSON files examined | 7,192 |
| Explicit complete tables recognized inside JSON | 85 |
| Additional graph classes found only in JSON | 4 |
| Combined spanning-root graph classes | 501 |
| Existing 194-state graph classes | 40 |

The large counts include raw compiler tables, experiments, rejected constructions, and small scanner fixtures. They are not counts of established consistency machines. Among the 40 classes at 194 states, 26 have a supplied ordinary PA argument and physical evidence, 10 are auxiliary or exploratory, and four

are explicitly rejected. These labels describe the supplied evidence; this project has not independently reproved every historical consistency claim.

The parser recognizes complete numeric binary five-column tables in `.tm`, `.txt`, `.tsv`, and `.csv` files, including nested ZIPs. The supplemental scanner recognizes explicit JSON action lists, maps, records, and embedded table strings, including signed movement encodings. It does not enumerate the outputs of every possible source generator. That limitation is recorded in the machine-readable census.

## 2.1 Principal design families

| Family | Main distinction | Comparison consequence |
|---|---|---|
| U208 | U209 parent machines with history-sensitive folds and several supported quotients. | Larger baseline; control-state folds require their own context proofs. |
| PA209–PA196 | James Chang's separate source and compiler lineage. | Important source interfaces; its folds are not interchangeable with U208's. |
| PA196 | Imported source, arithmetic calibration, and a larger axiom emitter. | Important logical and physical dependency; its one-way theorem alone is insufficient. |
| PA195 families | Separate or bundled diagonal heads; 68/7 or 67/6 reader budgets and controller variants. | Intermediate allocations and several genuinely different tables. |
| Primary PA194 | Separate diagonal head; 67 AX readers and seven compatible folds. | $134 + 67 - 7 = 194$. |
| PA194-68/8 and R | Alternative decoder associations and fold sets. | Same count; source identity and physical compatibility both matter. |
| C, CS, N, H, NS, HS | Cached results, reversed destructive comparisons, omitted MP pushes, or changed Shift tests. | Command-word compilers are controller-specific. |
| A, J, ZR, CZ, CSZ | Known-zero decrements used as returns or scheduler jumps. | Source savings need not delete a physical state. |
| D and Q | Change the diagonal head's parameter or exchange curried premises in quantifier distribution. | Require explicit proof translations. |
| M5/M6, N5/N6 | Change formula coefficients, optionally with no-push MP. | Require hereditary recoding and repair of numerical dominance. |
| Incoming QN, HA, HAZ, SUMS | New combinations and original-sum symmetric comparison. | Already part of the comparison baseline here. |

Two naming pitfalls are material. The placed cached-only NS machine is the same class as HS, whereas the incoming packet uses NS for a different original-sum design. Also, a no-push A/D-layout table exists only in `PA194NS_research.zip/audit/transition_graphs.json`; ignoring JSON would miss it. The complete 40-class table is Appendix A, and the 54-packet ledger is Appendix B.

## 2.2 Novelty of the delivered tables

The comparator checks complete canonical transition tuples, not merely hashes. For every existing 194-state table it tries all graph-spanning roots and both choices of tape reflection and bit complementation, canonically renaming the controls. It also ignores the final write and move into the halt label. A transformed positive control verifies that renaming, reflection, and complementation are actually

recognized. Both new tables have no match. This establishes the stated graph novelty; it does not establish semantic inequivalence, which would be contrary to the intended common halting condition.

The mathematical difference is explicit. M5 and M6 use the original quantifier-distribution head; the new machines use its premise-exchanged form. Existing Q and QN use coefficient four. PA194Q5 and PA194Q6 change the numerical constructors throughout the source and change the actual emitted axiom in its original stack slot. Their derivation languages and literal transition graphs therefore change, rather than just the names of the controls.

## 3 The revised numerical calculus

Let $p(a,b) = \frac{1}{2}(a+b)(a+b+1)+b$ be Cantor pairing. For $k = 5, 6$ set

$$\begin{aligned} I_k(A,B) &= kp(A,B), & M_k(a,b) &= kp(a,b)+1,\\ U_k(a,B) &= kp(B,a)+2, & E_k(a,b) &= kp(a,b)+3. \end{aligned}$$

Zero denotes truth and $1 = M_k(0,0)$ denotes contradiction. Negation is $N_k(B) = I_k(B,1)$; conjunction and existential quantification use their classical implicational definitions. The reversal of the arguments in $U_k$ is part of the construction. Unused residues are interpreted as truth in the arithmetic soundness translation.

Let $\mathcal{C}_k$ be the inherited separate-diagonal calculus, with its six ordinary heads, the dominated implicational head, the diagonal schema $\Gamma_k(B) = I_k(B, U_k(B,B))$, guarded pointed separation, modus ponens, and generalization. Let $\mathcal{C}_k^Q$ change only

$$\begin{aligned} h_{6,k}(D,A,H) &= U_k(D,A) \to \big(U_k(D, A \to H) \to U_k(D,H)\big)\\ \text{to } h^Q_{6,k}(D,A,H) &= U_k(D, A \to H) \to \big(U_k(D,A) \to U_k(D,H)\big). \end{aligned}$$

All arrows here are $I_k$. The parameter guard, other formulas, and positions in the emitted stack remain the same.

**Lemma 2** (Uniform premise exchange)**.** *There are primitive-recursive proof translations between $\mathcal{C}_k$ and $\mathcal{C}_k^Q$ preserving the literal conclusion. PA verifies their correctness.*

*Proof.* Use the finite implicational theorem

$$(P \to (Q \to R)) \to (Q \to (P \to R)).$$

The delivered certificate derives it from K, S, and modus ponens, without quantifier, negation, or falsity rules. Substituting $P = U_k(D,A)$, $Q = U_k(D, A \to H)$, and $R = U_k(D,H)$ gives one direction; interchanging $P$ and $Q$ gives the other. Replace each affected axiom leaf by this fixed derivation and leave all other derivation constructors unchanged. Induction on the derivation proves the translation claim.

K and S are uniformly derivable from the unchanged restricted implicational basis. Its proof requires reflexivity of numerical comparison and the two operand bounds $a, b \le I_k(a,b)$. These follow from $a, b \le p(a,b)$ for every $k \ge 1$. Consequently the same proof applies at five and six; one must not import only a coefficient-four conclusion without this check. Numerical aliases such as $I_k(0,0) = 0$ cause no problem: the certificate is instantiated as a homomorphism of implication terms and uses no syntactic injectivity. □

### 3.1 Why coefficient recoding preserves PA calibration

For completeness, the relevant inherited radix argument is stated here. Define $F_k(0) = 0$, and for $n = 4p(a, b) + r > 0$ put

$$F_k(n) = kp(F_k(a), F_k(b)) + r \qquad (0 \le r < 4).$$

Both recursive arguments are below $n$, so this is a primitive-recursive course-of-values definition. A primitive-recursive left inverse, decoding the first four residues and sending unused residues to zero, proves that $F_k$ is injective. It fixes 0 and 1 and commutes with all four constructors, *including the variable labels*. In particular,

$$F_k(\Gamma_4(B)) = \Gamma_k(F_k(B)).$$

Recoding formulas while leaving their numerical variable labels unchanged would fail at this diagonal schema.

There is a second obstacle: $F_k$ need not preserve order. For example, $40 < 44$ while $F_5(40) = 75 > 55 = F_5(44)$. The restricted propositional head has an inequality side condition, so a one-leaf translation is invalid. The inherited `rho8All` proof first derives the unrestricted head from the restricted one using only the operand bounds just stated. Each translated restricted leaf is replaced by that finite template. This produces a PA-verifiable translation of coefficient-four refutations into $\mathcal{C}_k$ refutations.

For the converse, interpret membership by finite binary membership, equality by equality, and $U_k$ by first-order universal quantification. The coding inequality proves that every variable label in a nonzero formula is smaller than its code. This supplies the diagonal axiom. PA's full induction proves finite binary comprehension for each arithmetic formula, supplying the pointed-separation instances. A fixed syntactic template produces the PA proof of each translated axiom. It handles the zero alias, unused residues, and coincident-label case separately. Thus a finite numerical refutation produces a finite PA refutation.

These are proof-producing transformations, not an assertion of a uniform truth predicate. The supplied ordinary calibration consequently gives

$$\mathrm{PA} \vdash (\exists d\ \mathrm{Prf}_{\mathcal{C}_k}(d, 1)) \leftrightarrow (\exists p\ \mathrm{Prf}_{\mathrm{PA}}(p, \ulcorner 0 = 1 \urcorner)).$$

The exchange lemma replaces $\mathcal{C}_k$ by $\mathcal{C}_k^Q$ in this equivalence. Full attributed manuscripts and the relevant generic Lean proof template are retained in `calibration/`; their status is ordinary inherited calibration, not a newly compiled endpoint.

## 4 Exact source and command compiler

The logical source register order is $(N, T, W, A, D, H, X, Y, L)$. Its physical register positions are $(6, 2, 0, 4, 3, 7, 8, 5, 1)$. The source instruction tree is parsed from the supplied PA196 program, with a new AX grammar and a single multiplication-body change.

The AX macro word still has 190 tokens. The only logical-head replacement is the half-open interval $[102, 121)$; all surrounding tokens are equal. At coefficient four, replacing the primary word by this Q word changes 79 resolved source instructions, at indices 1953 through 2067, out of 3,673. This check compares control-flow targets, not just the word's flattened operation names.

The constructor changes each of 61 distinct four-increment multiplication bodies into a five- or six-increment body. Every body is inside a verified Cantor constructor macro. The 3,429 remaining source sites agree under an explicit ordinal relocation. There are no external entries into the interior of an increment body. Therefore the complete lengths are

$$3673 + 61(5 - 4) = 3734, \qquad 3673 + 61(6 - 4) = 3795.$$

Free-algebra execution checks the entire AX effect: it pushes the incoming top, a generalization, the dominated head, six ordinary heads, and the diagonal head. Only pushed entry 8 in zero-based bottom-to-top order is changed by Q. The guarded top and final register cleanup are unchanged.

### 4.1 The controller retained here

The new machines use the *original* controller, whose Shift halts when $t + h = 1$ and whose successful MP retains a dummy zero on the encoded stack. The stack is Cantor encoded: $[a]w = p(a, w)$, with $p(0, 0) = 0$ explicitly allowed. This controller choice fixes the proof compiler; cached-controller command words cannot be substituted for it.

Writing stores as $(t, w, d, h)$, the ten-entry AX interface has the same safe extraction positions as the original separate-head construction. The following words and equations remain valid for $k = 5, 6$ and the Q head:

$$\begin{aligned} \mathrm{pop} &= \mathsf{AX}(0,0,0)\mathsf{S}^{11}, & \mathrm{pop}(t,[a]w,d,h) &= (t,w,0,a),\\ \mathrm{push} &= \mathsf{AX}(0,0,0)\mathsf{S}^{9}\mathsf{MP}(0), & \mathrm{push}(t,w,d,h) &= (0,[t]w,0,0),\\ \mathrm{discard} &= \mathsf{MP}(1)\mathsf{AX}(1,0,0)\mathsf{S}^{10}\mathsf{MP}(0), & \mathrm{discard}(t,w,0,0) &= (0,w,0,0). \end{aligned}$$

They also hold for $t = 1$: intermediate Shift tests in these words are protected by constructor residues and positivity. All six ordinary heads and $\Gamma_k$ are implication, quantifier, or existential codes; none is the contradiction code 1.

Let $q_1, \ldots, q_{10}$ be AX's saved entries in stack-top order:

$$\Gamma_k(A),\ h^Q_{6,k},\ h_5,\ h_4,\ h_3,\ h_2,\ h_1,\ \rho_{8,k},\ U_k(d,h),\ t.$$

For $1 \le j \le 7$, the word

$$\mathrm{discard}\,\mathsf{AX}(A,D,H)\mathsf{S}^{j+1}\mathrm{pop}^{9-j}$$

selects $q_j$ while preserving the protected stack and clearing both caches. The Q head is selected at $j = 2$, exactly the old sixth-head slot. Dominated-head and guarded-gate selectors are inherited unchanged.

Generalization uses the inherited offset word and $\mathrm{push\,offset}(i)\mathsf{SAX}(i,0,0)\mathsf{S}^{10}$. For MP the correct suffix is

$$\mathrm{suffix}_b = \mathsf{SSMP}(b)\mathsf{AX}(b,0,0)\mathsf{S}^{11}.$$

On $(0, [a][I_k(a, b)]w, 0, 0)$, it halts at the second Shift if $a = 1$. Otherwise MP yields $(b, [0]w, b, 0)$, and the remaining AX–Shift suffix returns $(b, w, 0, 0)$. This explicitly removes the original controller's dummy zero. It is the point at which a cached-controller proof would give the wrong command sequence.

Induction on a numerical derivation compiles it into a finite word $C(d)$ which, from every $(t, w, 0, 0)$, either halts or returns its conclusion with the protected $w$ and zero caches. A refutation gives the universally halting word $W_d = \mathsf{AX}(0,0,0)C(d)\mathsf{S}$. No reset of the actual store between candidate command words is assumed.

### 4.2 Enumeration and the reverse implication

The original command scheduler uses continuation-pointer encodings

$$\begin{aligned} e([]) &= 0, & e(\mathsf{S} :: v) &= 1 + p(0, e(v)),\\ e(\mathsf{MP}(x) :: v) &= 1 + p(1, p(x, e(v))), & e(\mathsf{AX}(c,x,y) :: v) &= 1 + p(c+2, p(y, p(x, e(v)))). \end{aligned}$$

Every positive residual pointer decreases under decoding. Thus each candidate finishes in a bounded number of commands or halts, and fresh candidates enumerate $2, 3, 4, \ldots$ from the verified initial seed.

The candidate coding of $W_d$ lies in that enumeration. Bounded induction in PA shows that it is reached or an earlier halt has occurred.

For soundness, annotate $t$, $h$, and the live *canonical* stack entries with PA proofs of their arithmetic translations. Popping zero supplies the truth proof; pushing a zero onto zero adds no canonical cell. AX, successful or failed MP, and nonhalting Shift preserve the annotation. If $t + h = 1$, one of the two annotated formulas is code 1; its translation is refutable in PA, yielding a PA contradiction. These syntactic annotation operations and their correctness induction are primitive recursive and PA-verifiable.

Consequently the source halts if and only if there is a PA refutation, provably in PA. Section 5 connects that source to the literal 194-state tables.

# 5 The 202-to-194 physical construction

The frozen Q grammar allocates 67 AX reader states. The fixed controller has 135 controls after adding one reader for three increments. In native labels, the constructor changes are

| | 0 | 1 | |
|---|---|---|---|
| 190 | $(0, R, 330)$ | $(1, R, 99)$ | $(k = 5)$ |
| 190 | $(0, R, 330)$ | $(1, R, 330)$ | $(k = 6)$ |
| 330 | $(0, R, 61)$ | $(1, R, 99)$. | |

The complete rows, including all unchanged controller and arithmetic rows, are reconstructed from the included inputs. The new three-increment reader permits a further context fold between native states 41 and 99, in addition to seven AX-reader folds. The count is therefore

$$\underbrace{135}_{\text{fixed}} + \underbrace{67}_{\text{AX readers}} - \underbrace{8}_{\text{simultaneous folds}} = 194.$$

| Deleted | Retained | New branch 0 | New branch 1 |
|---|---|---|---|
| 2004 | 2052 | 0L2015 | 1L2048 |
| 2021 | 2030 | 0R90 | 1L2023 |
| 2025 | 2035 | 0L2002 | 1R61 |
| 2027 | 2046 | 0L2036 | 1R2031 |
| 2039 | 2044 | 0R2040 | 1L2014 |
| 2055 | 2056 | 0L2042 | 1R2043 |
| 2058 | 2017 | 0L2050 | 1L2062 |
| 41 | 99 | 0L30 | 1L2040 |

All labels in this fold table are native compiler labels; the delivered tables use a dense renumbering.

## 5.1 Why the folds preserve executions

Let $\alpha$ identify each deleted control $a$ with its retained partner $b$. Each arrival at $a$ has just moved right leaving a 0 behind; each arrival at $b$ has left a 1. That predecessor bit distinguishes the two old roles. For each pair, the checker considers both roles and both current read bits, verifies all incoming transitions, and executes the *final simultaneously folded table*. It restores the two-cell tape and reaches exactly the old transition's projected endpoint in a positive number of moves. None of the eight folds touches the start or halt.

All other transitions commute directly with $\alpha$. Induction on unfolded steps, concatenating these local positive traces, proves the forward simulation. Determinism and the absence of internal halts in a replacement trace give the finite-halting equivalence from admissible starts. Checking folds one at a time against different intermediate tables would not establish this simultaneous result.

### 5.2 Primitive contracts and initialization

A source frame contains a source path, an arbitrary affine zero gap, the nine natural-valued registers in the stated physical order, and an arbitrary finite nonempty spare list. The header parameter is 48; the maximum path length here is 35. Each increment, both branches of every decrement, every jump, and the halt leaf is checked at its resolved source address. The two accepted tables leave all inherited arithmetic-box transition rows unchanged.

The construction checker passes 4,841 and 4,902 universal symbolic contracts, respectively. The totals 1,546,445 and 1,571,284 in its receipts count executor operations with symbolic loop/box summaries; they are *not* running-time bounds on arbitrary natural-valued registers. A second implementation parses the source independently, resolves its control flow by two methods, and checks literal-tape behavior using list lemmas rederived from the actual transition rows.

| Independent literal-tape check | PA194Q5 | PA194Q6 |
|---|---:|---:|
| Primitive outcomes | 4,841 | 4,902 |
| Symbolic executor steps | 1,131,298 | 1,149,494 |
| Maximum symbolic steps per outcome | 397 | 397 |
| Minimum positive-duration constant | 3 | 3 |
| Generic list lemmas from candidate rows | 4 | 4 |
| Arithmetic-box oracle used | No | No |

The blank-start check is an exact finite computation of 58,320 moves. It compares the entire set of nonzero cells, the native control 84, and head position $-2$ with the expected source-site-25 frame. Its logical register vector is $[2, 0, 0, 0, 0, 0, 0, 0, 0]$ and it has 26 spare blocks of value 1. The exterior is all zero. Thus a successful check is stronger than merely observing a plausible register prefix or a long nonhalting run.

The arithmetic-loop lemmas are proved by induction on unary lengths and finite lists. Source paths, frame encodings, table lookup, finite traces, and trace concatenation are primitive recursive. PA can formalize these inductions and the candidate-specific finite checks. Every primitive simulation has positive physical duration. Under the annotated-source invariant, $T + 1$ successive primitive boundaries therefore extend past any alleged physical halting time $T$, contradicting determinism. In the other direction, a compiled source refutation yields a finite literal halting trace. This proves the required physical/source transport and completes the stated PA equivalence.

## 6 Unsuccessful reductions and the useful remaining obstacles

The bounded searches in this session are recorded in `search/summary.json`. They explore premise exchanges in the third, fourth, and sixth ordinary heads, alone and in combinations, and decoder reassociation with the changed radix controller and the extra 41/99 fold. Their stopping conditions are finite time budgets. Their best allocation scores did not yield a verified table below 194. These observations imply no lower bound on the number of states. Ten complete allocation runs performed 14,425,456 mutation iterations; the four coefficient-five/six runs retained best projected state count 194. The frozen accepted constructions do not depend on repeating a stochastic search.

One source experiment moves the contradiction test from Shift to the MP operation which first creates code 1. The first-production argument is mathematically plausible because AX and generalization do not first produce 1. An adapted ZFC context allocator reproduced the old 194-state positive control, but its changed 208-state compilation failed a primitive contract. That table is explicitly rejected. Decoder reuse must respect the interpreter's later passes, not just its first arrival at a source leaf.

An additional physical investigation uses the existing PA194QN table. It derives 3,236 must-zero facts, 5,924 must-positive facts, and 750 nontrivial modulo-four facts at source sites. The exact blank

startup also supplies the invariant that every spare register equals one. Both completed collectors cover all 4,755 primitive outcomes and all 194 controls. These stronger invariants reduce the radius-five context set from 29,479 to 24,388 windows, removing 5,091 and adding none.

Both context sets were screened for common-action state mergers, per-incoming-edge bypasses, and a new predecessor-history-marking fold. The latter lets predecessors overwrite their cells to distinguish two roles, requiring the detour to restore the complete local tape. It tests 14,401 independent control pairs and 22,148,352 candidate first-action trials per context set. All six screens completed at a 300-step detour bound and returned zero candidates. The supplied inputs and receipts bind these negative results to QN, not to Q5 or Q6. They exclude only the stated finite families, not arbitrary smaller machines.

A guard-free logical proposal was also reviewed. Taking a separator label $z = M_4(D, I_4(H, A))$ guarantees $z > D, H, A$, including the all-zero case. This establishes arithmetic soundness of the suitably split separator/property gate. It does not supply the arbitrary-label membership-vacuity instances required by the present completeness proof. Using full variable-renaming rules at that point would assume an interface which the small calculus has not yet derived. The proposal therefore remains a proof obligation, not an accepted consistency construction.

The ZFC material contributes reusable decoder allocation and local simulation methods, but its consistency strength cannot replace PA's. Similarly, replacing the source by a small iPRA checker would search for iPRA refutations. A route through intuitionistic arithmetic would need full induction, a syntactic PA/HA refutation translation, and its own complete machine compilation.

## 7 Reproducibility and proof status

Run `python3reproduce.py` from the archive root with Python 3.10 or later, with assertions enabled. It reconstructs both tables, checks the source and physical certificates, replays premise exchange, runs the independent audit, and checks graph novelty. No network access, external repository, Lean installation, or stochastic rediscovery is required. The optional discovery programs require a C++17 compiler. To rebuild this report, change to the `report/` directory and run `pdflatex` twice on `pa194_qradix.tex`.

The archive retains exact tables, source trees, resolved instructions, control-label bijections, fold traces, startup supports, verification receipts, comparison records, inherited mathematical references, licenses, and optional search inputs and logs. The main construction is deterministic; wall-time-limited discovery searches need not rediscover the same grammar.

There are three separate assurance levels. The exact files and executable certificates have been checked in this session. The PA-provability claim has an ordinary mathematical proof using the stated inherited proof templates. The complete claim has not been newly checked in a proof assistant. The repository also explicitly distinguishes its unfinished iPRA internalization from its standard finite-run Lean developments; this report does not collapse those distinctions.

## 8 Further research questions

1. Can the Q/radix decoder use 66 AX readers with eight compatible folds, or 67 readers with nine, while retaining the same 135 fixed controls? Either would give 193 states. All folds must be checked in the final table, including the fixed-controller pair.

2. Can the proven source invariants and one-valued spare-list language justify a physical state elimination outside the completed two-control screens? A positive answer needs an invariant closed under the actual blank-start computation, not a finite simulation sample.

3. Can contradiction testing be moved to successful MP using a compiler that preserves the interpreter's multi-pass source-prefix invariants and reconstructs a valid blank seed?

4. Can higher-order predecessor histories share three or more reader roles with a smaller compatible quotient, avoiding the limitations of disjoint two-role folds?
5. Can a uniformly fresh separator construction recover the arbitrary-label vacuity interface without presupposing full first-order renaming? Resolving that logical obstruction could make guard removal a valid source-size reduction.
6. Can the ordinary PA calibration and these explicit physical certificates be internalized together in Lean/Rocq, and then at the repository's stronger iPRA target? This is independent of whether the literal state count drops below 194.

## Sources and attribution

All repository citations are to the fixed ProveIt commit `317b07268681b44ff352d173a162e2a8c241b2fe` unless an included historical document explicitly states its own earlier provenance.

[1] PA overview and delivered-machine audit: https://github.com/VladimirReshetnikov/ProveIt/tree/317b07268681b44ff352d173a162e2a8c241b2fe/Computability/BusyBeaver/Research/PA.

[2] Primary separate-head PA194 construction and physical certificate: `PA/docs/pa194_conpa_2026-11-13/` within that tree.

[3] Original ten-entry command compiler and PA calibration: `PA/docs/pa194_conpa_2026-11-13_2/mathematics/pa194_calibration.tex`.

[4] Uniform coefficient recoding, source-binding proof, and 41/99 fold: `PA/docs/pa194_radix_family_2026-11-13/`.

[5] Q head, implicational certificate, and independent tape engine: `PA/docs/pa194q_conpa_2026-11-13/`.

[6] James Chang's imported PA196 source and generic restricted implication/runtime proofs: `PA/jamesyc-upstream/pa196/`.

[7] Current incoming packets: https://github.com/VladimirReshetnikov/ProveIt/tree/317b07268681b44ff352d173a162e2a8c241b2fe/docs/incoming.

[8] ZFC compiler ideas and the ConciseIPRA boundary: `Computability/BusyBeaver/Research/ZFC/` and `Logic/PrimitiveRecursiveArithmetic/ConciseIPRA/`.

Imported code and mathematical components retain their attribution and license notices. The new work is the combined Q5/Q6 construction and its integration checks, the updated literal/JSON comparison, and the bounded experiments and proof reviews documented here.

# A All 40 existing 194-state classes

This table classifies the prior corpus, excluding the two new machines. "Ordinary PA argument" reports supplied manuscript evidence, not a fresh proof-assistant verification. Exact paths, hashes, graph identifiers, and representative tables are included in `comparison/`.

| Design | Distinguishing mechanism | Status |
|---|---|---|
| CS both-zero6 | Known-zero X decrement at both source26 and final emitter exit; research composition. | Auxiliary |
| CSZ-A3 | CSZ plus comparator-return register3 and final-emitter register3 decrements. | Auxiliary |
| CSZ-A5 | CSZ plus comparator-return register3 and final-emitter register5 decrements. | Auxiliary |
| CSZ-A6 | CSZ plus comparator-return register3 and final-emitter register6 decrements. | Auxiliary |
| Cached I-tree zero-jump seed | Alternate decoder plus cached controller and entry zero jump; search basis. | Auxiliary |
| Gamma(D)-CSZA | Supplementary Gamma(D) emitter plus known-zero changes. | Auxiliary |
| M4R | Reassociated coefficient4 constructor decoder; original resolved source graph. | Ordinary PA argument |
| M5 | Formula constructor coefficient5; explicit variable and proof recoding. | Ordinary PA argument |
| M6 | Formula constructor coefficient6; explicit variable and proof recoding. | Ordinary PA argument |
| MPH | MP result cached in H while retaining original Shift; physical search input. | Auxiliary |
| PA194 | Separate diagonal head, 67-reader decoder with seven folds, original controller. | Ordinary PA argument |
| PA194-68/8 | Same separate-head source, reassociated 68-reader decoder with eight folds. | Ordinary PA argument |
| PA194-AD | PA194 with physical A/D register exchange. | Ordinary PA argument |
| PA194A | CS entry jump at source26 becomes decrement of known-zero register A. | Ordinary PA argument |
| PA194C | MP result remains cached in H; Shift tests H=1; original ten-entry logical interface recompiled. | Ordinary PA argument |
| PA194CS | Symmetric Shift guard; cached MP controller; contextual controller fold. | Ordinary PA argument |
| PA194CSZ = PA194X | CS entry jump at source26 becomes decrement X; two headline names denote identical dense bytes. | Ordinary PA argument |
| PA194CZ | Cached controller; entry known-zero decrement of register 6. | Auxiliary |
| PA194D | Separate diagonal axiom uses Gamma(D), new proof selector, 69 readers and nine folds. | Ordinary PA argument |
| PA194F | CS final emitter jump becomes decrement of known-zero register X. | Ordinary PA argument |
| PA194H | No-push MP returns result in visible top; Shift tests cached H; original comparison implementation. | Ordinary PA argument |
| PA194HA | MP result cached in H with original Shift; A/D physical register placement. | Ordinary PA argument |
| PA194HAZ | HA plus a proved known-zero decrement at the final AX return. | Ordinary PA argument |

| Design | Distinguishing mechanism | Status |
|---|---|---|
| PA194HS = PA194NS | No-push MP returns result in visible top; Shift tests cached H with symmetric comparison. Two deliveries have identical dense bytes. | Ordinary PA argument |
| PA194J | Ten source jumps become known-zero decrements; three physical rows change. | Ordinary PA argument |
| PA194N | Successful MP drops dummy-zero push, retains original Shift; revised command proof compiler. | Ordinary PA argument |
| PA194N5 | Coefficient-five formula constructors plus no-push MP; original Shift. | Ordinary PA argument |
| PA194N6 | Coefficient-six formula constructors plus no-push MP; original Shift. Two deliveries have identical dense bytes. | Ordinary PA argument |
| PA194NS (sum guard) | SUMS plus no-push MP; different from the placed cached-only NS=HS. | Ordinary PA argument |
| PA194N_AD (JSON only) | No-push MP with original sum Shift and A/D physical register placement. | Auxiliary |
| PA194Q | Quantifier-distribution curried premises exchanged; K/S proof translation. | Ordinary PA argument |
| PA194QN | Q quantifier-distribution premise exchange plus N no-push MP and original Shift. | Ordinary PA argument |
| PA194R | Reassociated decoder of PA194F; 1111 source paths changed, same resolved source graph. | Ordinary PA argument |
| PA194SUMS | Original sum Shift guard with reversed destructive comparison; contextual eighth fold. | Ordinary PA argument |
| PA194ZR | Eight comparator returns and final emitter return become known-zero register3 decrements. | Ordinary PA argument |
| PA194altC | Cached controller on alternate 68-reader decoder. | Auxiliary |
| Rejected CS node1-X | Native103/read1 becomes dec6; exact blank run halts at1361. | Rejected |
| Rejected I-tree node1-X | Known-zero source rewrite fails blank bootstrap; exact halt at1361. | Rejected |
| Rejected ZR node1-X | Register6 comparator-return variant; exact blank run halts at1361. | Rejected |
| Rejected seven-zero scanner | Decoder local folds pass, but source-address scanner violates runtime contract. | Rejected |

# B The 54-packet comparison ledger

The full machine-readable ledger includes fixed-commit source URLs and duplicate notes. This compact version gives each packet's main role.

| Packet | Principal contribution and qualification |
|---|---|
| `Below_U208_Research_2026-10-29` | Source-boundary guards, displaced-head quotient lemma, closed-loop pruning, and broader negative local searches. Pruning 3909 to 3904 source occurrences increases the shared-tree cost. No accepted 207. |
| `ConPA_below208_research_2026-10-29` | Two-object obstruction to collapsing source/output cover roles, exhaustive restricted axiom-parameter permutation search, two-row and register-invariant searches. No accepted 207. |
| `PA194R_research_2026-11-14` | R changes the decoder association; F changes a final known-zero return; A changes a zero-entry return. The packet's X table duplicates CSZ. All remain at 194 states. |
| `PA195_ConPA_research_2026-11-14` | 195S uses 67 AX readers and six folds with the separate-head source unchanged (134+67-6). 195C retains that grammar with the cached controller. Both receive complete symbolic and independent physical audits in the packet. |
| `PA_consistency_TM_research_2026-11-13` | New source-only zero-run-five witness (3812 sites,159 internal nodes,depth37), combined-negation/MP3573-site157-node witness, and parameter-free residual MP. Fixed/protected physical allocations remain rejected; many bounded quotient searches are negative. |
| `U208_ConPA_2026-10-29` | Complete U208, original class A: identify old native roles 172 and 211, with helper 217 (or 214 under alternate native labeling). The local simulation is a finite all-tape argument; full PA calibration is inherited and written, not newly kernel exported. |
| `U208_ConPA_Conditional_Context_2026-10-29` | Same conditional-history reduction; primary and alternate presentations give the two original U208 graph classes. Four local blocks have lengths 7,1,3,1. |
| `U208_ConPA_Entry_Context_2026-10-29` | Re-expresses the same reduction by entry guards, with primary and compatibility alternatives. Its saving is not additive with the previous packets. |
| `U208_ConPA_History_2026-10-29` | Historical-label presentation of the same two classes and same one-state deletion; includes a failed further reduction. |
| `conpa_tm_comparison_and_reduction_search` | Two experimental196 MP commands (result in H, omit zero push), with mathematical command arguments but only bounded physical frame evidence. Two rejected Shift layouts actually halt from blank after43563 steps. |
| `conpa_tm_comparison_and_search` | Accepted AD-register-swap196. Two-rotation195 is rejected by 93 primitive-contract failures. Source geometry changes cannot reuse old contextual folds automatically. |
| `conpa_tm_study_2026-11-13` | Combined196 implements H-only Shift and MP-in-H, with4928 uniform contracts; an additional rebracketed Combined196 has full runtime and startup confirmed in this audit; direct-negation197 has3556 sites. A157-node source receives a larger207-state realization. |
| `pa194C_conpa_2026-11-13_2` | C moves successful MP's result to cached H and makes Shift test H alone. It changes the command compiler, not only physical state names. This same C graph appears in five principal deliveries; auxiliary layouts are separately inventoried. |
| `pa194N_conpa_2026-11-13` | N removes the successful-MP dummy-zero push while retaining the original sum test in Shift. Its all-store proof compiler inserts a reserved zero in the proof word; simply deleting the old push without changing the compiler would be incorrect. |
| `pa194_conpa_2026-11-13` | Separate-head 190-token emitter with 67 reader states and seven contextual folds: 134+67-7=194. Exact source/CFG/physical startup and ordinary PA calibration are supplied. |
| `pa194_conpa_2026-11-13_2` | Alternative separate-head layout uses 68 readers and eight folds: 134+68-8=194. Includes original-controller all-store proof compiler and an independent free-algebra emitter/command audit. |
| `pa194_new_designs_2026-11-13` | The C cached-result/cached-only-Shift table is repeated elsewhere; the additional CS machine symmetrizes Shift comparison. The physical tables receive their own changed-command calibration. |
| `pa194_radix_family_2026-11-13` | M5 and M6 replace formula multiplier 4 by 5 or 6; hereditary formula/label recoding, unrestricted rho8 derivation, and all-residue PA interpretation calibrate them. M4R is a distinct reassociation at radix four. |
| `pa194c_conpa` | C moves successful MP's result to cached H and makes Shift test H alone. It changes the command compiler, not only physical state names. This same C graph appears in five principal deliveries; auxiliary layouts are separately inventoried. |
| `pa194c_conpa_2026-11-13` | C moves successful MP's result to cached H and makes Shift test H alone. It changes the command compiler, not only physical state names. This same C graph appears in five principal deliveries; auxiliary layouts are separately inventoried. |

| Packet | Principal contribution and qualification |
|---|---|
| pa194c_conpa_2026-11-13_1 | C moves successful MP's result to cached H and makes Shift test H alone. It changes the command compiler, not only physical state names. This same C graph appears in five principal deliveries; auxiliary layouts are separately inventoried. |
| pa194hs_conpa_2026-11-13 | PA194H and PA194HS; successful MP omits dummy-zero push and leaves result in visible T; Shift tests cached H. HS uses symmetric comparison. Both 194 states. |
| pa194j_pa194d_conpa_2026-11-13 | J replaces ten proved-zero jump returns by decrement returns. D constructs the diagonal head from a different source parameter. Their source commands differ materially; both remain 194. |
| pa194n5_n6_conpa_2026-11-13 | PA194N5 and PA194N6; coefficient-5/6 formula constructors combined with no-push MP. Two 194-state machines. Node verifier supplied. |
| pa194n6_research_2026-11-13 | PA194N6; coefficient-six formula constructors combined with no-push MP, 3776 sites and 4877 primitive outcomes, independently audited literal runtime. |
| pa194ns_conpa_2026-11-13 | PA194NS; no-push MP and symmetric cached-head Shift, with all-store compiler and 4750 primitive outcomes. No machine below 194. |
| pa194q_conpa_2026-11-13 | Q exchanges the two curried premises of the sixth quantifier head using a finite implicational derivation. CSZ combines symmetric Shift with a known-zero entry. Q's controller is cached-only; CSZ is later duplicated as X. |
| pa194zr_conpa_2026-11-13 | ZR uses nine known-zero return replacements in a new layout. Its additional node1-X edit is rejected by a 1361-step blank halt. |
| pa195_conpa_2026-11-13 | The bundled diagonal-vacuity calculus receives a 195-state allocation (68 AX readers, seven folds). Its written calibration supplies the syntactic calculus-to-PA direction beyond the upstream one-way internal theorem. |
| pa195_diagonal_combined_2026-11-13 | PA195DC combines the bundled diagonal emitter with the Combined controller; six native rows change, with 3676 source sites and complete contracts/startup. Two secondary 195-state allocations are separately marked experimental. |
| pa195_followup_research | M195S8 is a supported 195-state eight-fold layout. The tempting seven-zero scanner 194 quotient passes its local folds but fails a primitive source contract; it is expressly rejected. |
| pa196_research_2026-11-13 | Exact 157-node representation of the original3812-site source, down from159; seed stays at source path100000. Claimed prospective195 allocation loses a required decoder helper. Native HA/full-induction theory is a separate research direction. |
| pa208_research_2026-10-29 | Eighteen additional pairwise nonisomorphic U208 helper variants, each with a local halting-equivalence argument. Expands accepted U208 corpus to 20 classes; no accepted 207. |
| pa_below196_research | Eight equivalent gate spellings, orientation involutions, exact freshness obstruction R(5)=9, smaller pairing macro that increases DAG cost, H-only guard specialization, and broader finite physical searches. No new accepted smaller table. |
| pa_consistency_comparison_2026_11_13 | First quantifier head split into modal B plus explosion, with finite Hilbert certificates in both directions. Countermodels obstruct deleting either membership rewrite head or replacing the first quantifier head by reflexivity. Twelve source/table combinations retain separate command/calibration obligations. |
| pa_consistency_comparison_and_new196_2026-11-13 | Three accepted diagonal-calculus196 layouts: diagonal196 (191 AX items,3700 sites), separate196 (190 items,3673 sites), scanner_best196 (same3673 sites, improved address layout). Also a runtime/startup-checked scanner196 on original source and four auxiliary diagonal closing-CD variants with differential runtime proofs, trit invariants, and exact startup. Short-scanner195 and U208/PA196 hybrid startup attempts fail stated obligations. |
| pa_consistency_cross_archive | Four supported 196-state compositions: bundled/separate diagonal emitter, each with Combined controller and with/without zero-jump closing. They retain exact source/runtime/startup evidence. Direct negation gives a separate 197-state composition; automatic freshness remains incomplete. |
| pa_consistency_reduction | Rebracketed196; H-only Shift196 with3811 sites and57922-step startup; integrated witness gate has3745 sites but best physical table203. Two optimistic195 allocations fail, one by an exact short cycle on a source frame. |
| pa_consistency_reduction_audit_2026-11-14 | Comparison and bounded reduction audit. Explicitly no accepted sub-196 machine or new end-to-end theorem; includes scanner-synthesis positive controls that must not be counted as PA machines. |
| pa_consistency_research | New158-node source and an accepted196 regrouping; the eighth scanner zero consumes the apparent one-state saving. Its seven-zero195 version misroots on a concrete source frame. |
| pa_consistency_research_2026-10-29 | Proves zero-guarded source specialization 3909 to 3904 instructions and preserves the ordinal control graph; no smaller physical table. |

| Packet | Principal contribution and qualification |
|---|---|
| `pa_consistency_research_2026-11-13` | Diagonal vacuity B→U(B,B), exploiting formula-code freshness; AX201 to191, source3812 to3700. Initial complete implementation199. Gives detailed written reverse PA proof translation; no new full Lean endpoint. |
| `pa_consistency_research_2026-11-13_2` | New twelve-rotation196 and diagonal198. Also a command calculus halting at successful MP deriving code1; no new binary table for that command change. |
| `pa_consistency_research_2026-11-14` | NEG18 implements encoded negation using INC R7 twice then node64. Eight replacements cut 3673 source sites to 3441, but the physical allocation is 134+68-5=197. Full scratch invariant, 4468 contracts, and exact startup are supplied. |
| `pa_tm_comparison_and_reductions` | Two accepted196 rotations. Equivalent gate reordering shortens AX201 to199 macro items and source3812 to3773, but best physical compilation198. Finite free-algebra/logic certificates and restricted grammar searches are retained. |
| `pa_tm_extended_research_2026-11-13` | Four distinct one-row closing-CD variants remain196, with constrained symbolic runtime and unchanged blank startup. Joint source invariants and a generalized register executor strengthen search. |
| `pa_tm_research_2026-11-13` | New alternative196 with head order (1,2,3,5,6,4), unchanged 3812-site CFG, maximum source depth 36 rather than 38. Physical Lean baseline rebuild is supplied. R6-pruned198 passes runtime but lacks calibrated blank startup. |
| `pa_tm_research_2026-11-13_2` | New zerojump196 and an auxiliary CD3 parent (fully replayed in this audit): native27/1 changes 0 L40 to1 R21 under the source26 R6=0 invariant; startup58402. Deleting the resulting prospective helper gives a195 table with wrong source successor and R6 increment, witnessed also on the actual blank execution. |
| `pa_unary_power_cover` | A unary-matrix power-cover theory and written mutual consistency calibration with full PA. Supplies numerical proof-search interface and logical separations; no new literal TM. |
| `u208_comparison_research_2026-10-29` | Exact comparison, source/encoding research and a fully specified rejected207 frame witness. Weak/exact schema equivalence is separate from the still-open runtime/calibration replacement. |
| `u208_research_2026-10-29` | Both original U208 classes with symbolic runtime and startup evidence; no further state saving. |
| `PA194NS_research` (incoming) | PA194SUMS and another PA194NS preserve the original sum Shift guard but use symmetric destructive comparison; NS additionally omits successful-MP zero push. A JSON-only N_AD candidate is auxiliary. |
| `pa194haz_research_2026-11-14` (incoming) | PA194HA and PA194HAZ combine cached-result MP with original Shift and A/D physical placement; HAZ additionally replaces final AX return with a proved zero decrement. |
| `pa194qn_research_2026-11-13` (incoming) | PA194QN combines the Q premise-exchanged quantifier emitter with N no-push MP and original sum Shift; two 195-state tail variants remain experiments. |

# C Complete transition tables

An entry $wDq$ means write bit $w$, move in direction $D$, and enter $q$. The symbol H denotes halt label 194. Each row below displays two working states; both read-bit branches are given for each. The five-column `.tm` files are the canonical machine-readable deliverables.

## C.1 PA194Q5

| State | Read 0 | Read 1 | State | Read 0 | Read 1 |
|---|---|---|---|---|---|
| 0 | 1R1 | 1R2 | 97 | 0R117 | 1R97 |
| 1 | 1R3 | 1R1 | 98 | 0R30 | 1R49 |
| 2 | 0R0 | 1R2 | 99 | 0R48 | 1R61 |
| 3 | 0R4 | 0R1 | 100 | 0R87 | 1L118 |
| 4 | 1R5 | 1L6 | 101 | 0R106 | 0L14 |
| 5 | 1L7 | 1L8 | 102 | 0R36 | 1R119 |
| 6 | 0L7 | 1L6 | 103 | 0R118 | 1R79 |
| 7 | 0L9 | 1L6 | 104 | 0L46 | 0R120 |
| 8 | 0L10 | 0L11 | 105 | 0L45 | 1R121 |
| 9 | 1R12 | 0L9 | 106 | 0L96 | 1R122 |
| 10 | 0L9 | 0L10 | 107 | 0R123 | 0L14 |
| 11 | 1L8 | 1L11 | 108 | 0R72 | 1R108 |
| 12 | 0R12 | 0R13 | 109 | 0R120 | 1R109 |
| 13 | 0L14 | 0R13 | 110 | 0L46 | 0R108 |
| 14 | 1L15 | 0L15 | 111 | 0R71 | 0R97 |
| 15 | 0L15 | 1L16 | 112 | 0R59 | 0R97 |
| 16 | 0L17 | 1L16 | 113 | 0L65 | 1R124 |
| 17 | 0L18 | 1L16 | 114 | 0R125 | 1R126 |
| 18 | 0L19 | 1L16 | 115 | 0R109 | 1R115 |
| 19 | 0L20 | 1L16 | 116 | 0R127 | 1R128 |
| 20 | 0L21 | 1L16 | 117 | 0R115 | 1R117 |
| 21 | 0L22 | 1L16 | 118 | 0R129 | 1L67 |
| 22 | 0L23 | 1L16 | 119 | 0R130 | 1R131 |
| 23 | 0R23 | 1R24 | 120 | 0R14 | 1R120 |
| 24 | 0R25 | 1R26 | 121 | 0R94 | 0L14 |
| 25 | 0R27 | 1R28 | 122 | 0R96 | 1L100 |
| 26 | 0R29 | 1R30 | 123 | 0R132 | 1R48 |
| 27 | 0R31 | 1RH | 124 | 0R133 | 1R134 |
| 28 | 0L32 | 1R33 | 125 | 0R32 | 1R36 |
| 29 | 0R34 | 1R35 | 126 | 0R31 | 1R80 |
| 30 | 0L36 | 1L37 | 127 | 0R135 | 1R136 |
| 31 | 0R38 | 1R39 | 128 | 0R137 | 1R138 |
| 32 | 0R40 | 1R41 | 129 | 0R59 | 0R117 |
| 33 | 0R42 | 1R43 | 130 | 0R104 | 1R121 |
| 34 | 0R44 | 1R45 | 131 | 0R83 | 1R82 |
| 35 | 0R46 | 1R47 | 132 | 0R94 | 1R48 |
| 36 | 0R48 | 1R49 | 133 | 0R138 | 1R139 |
| 37 | 0R50 | 1R51 | 134 | 0R140 | 1R141 |
| 38 | 0R52 | 1R42 | 135 | 0R102 | 1R142 |
| 39 | 0R53 | 1R54 | 136 | 0L138 | 1R94 |
| 40 | 0R55 | 1R56 | 137 | 0R41 | 1L143 |
| 41 | 0R57 | 1R58 | 138 | 0R144 | 1R145 |
| 42 | 0R59 | 0R60 | 139 | 0R146 | 1R147 |
| 43 | 0R61 | 0L14 | 140 | 0R148 | 1R91 |

| State | Read 0 | Read 1 | State | Read 0 | Read 1 |
|---|---|---|---|---|---|
| 44 | 0R62 | 1R63 | 141 | 0R149 | 0L10 |
| 45 | 0R55 | 1R64 | 142 | 0R91 | 1R64 |
| 46 | 0L65 | 1R66 | 143 | 0R148 | 1R145 |
| 47 | 0R36 | 1R67 | 144 | 0R102 | 1R67 |
| 48 | 0L68 | 1L69 | 145 | 0R150 | 1R48 |
| 49 | 0R70 | 0L14 | 146 | 0R151 | 1R152 |
| 50 | 0R71 | 0R72 | 147 | 0R153 | 1R154 |
| 51 | 0R73 | 0L49 | 148 | 0R136 | 1R41 |
| 52 | 0R74 | 1R54 | 149 | 0R155 | 1R156 |
| 53 | 0L75 | 0L15 | 150 | 0L157 | 1L153 |
| 54 | 0R76 | 0L10 | 151 | 0R28 | 1R158 |
| 55 | 0R77 | 1R78 | 152 | 0R145 | 1R159 |
| 56 | 0R79 | 1R80 | 153 | 0R137 | 1R160 |
| 57 | 0R81 | 1R82 | 154 | 0R161 | 1R162 |
| 58 | 0R83 | 1R84 | 155 | 0R31 | 1R53 |
| 59 | 0L85 | 0R1 | 156 | 0R163 | 1R155 |
| 60 | 0R86 | 1R60 | 157 | 0R164 | 1R165 |
| 61 | 0L77 | 1L68 | 158 | 0R166 | 1R167 |
| 62 | 0R87 | 0L7 | 159 | 0L167 | 1R168 |
| 63 | 0R88 | 1R89 | 160 | 0R169 | 1R170 |
| 64 | 0R42 | 1R90 | 161 | 0R171 | 1R171 |
| 65 | 0R91 | 1R49 | 162 | 0R172 | 1R173 |
| 66 | 0R92 | 0L76 | 163 | 0R98 | 1R118 |
| 67 | 0R68 | 1R33 | 164 | 0R80 | 1R172 |
| 68 | 0R74 | 1R93 | 165 | 0R135 | 1R172 |
| 69 | 0R94 | 1R95 | 166 | 0R174 | 1R175 |
| 70 | 0L96 | 1L33 | 167 | 0R176 | 1R165 |
| 71 | 0L85 | 0R2 | 168 | 0R145 | 1R138 |
| 72 | 0R97 | 1R72 | 169 | 0R177 | 1R178 |
| 73 | 0R76 | 1R98 | 170 | 0R179 | 1R178 |
| 74 | 0R59 | 0R86 | 171 | 0L116 | 1R127 |
| 75 | 0R68 | 1R99 | 172 | 0L169 | 1L180 |
| 76 | 0R42 | 0L13 | 173 | 0R143 | 1R180 |
| 77 | 0R87 | 1R37 | 174 | 0R144 | 1R181 |
| 78 | 0R30 | 1R100 | 175 | 0R159 | 1R170 |
| 79 | 0R42 | 1R101 | 176 | 0R182 | 1R183 |
| 80 | 0R102 | 1R103 | 177 | 0R135 | 1R150 |
| 81 | 0R104 | 1L105 | 178 | 0R144 | 1R136 |
| 82 | 0R30 | 1R81 | 179 | 0R184 | 1R41 |
| 83 | 0R99 | 1R106 | 180 | 0R185 | 1R179 |
| 84 | 0R30 | 1R107 | 181 | 0R186 | 1R172 |
| 85 | 1R85 | 1L15 | 182 | 0R144 | 1R184 |
| 86 | 0R108 | 1R86 | 183 | 0R187 | 1R187 |
| 87 | 0L46 | 0R109 | 184 | 0R135 | 1R137 |
| 88 | 0R110 | 1R100 | 185 | 0R171 | 1R188 |
| 89 | 0R110 | 1R70 | 186 | 0R150 | 1R189 |
| 90 | 0R111 | 0L14 | 187 | 0R69 | 1L190 |
| 91 | 0R112 | 1L75 | 188 | 0R191 | 1R41 |
| 92 | 0R113 | 1R114 | 189 | 0R32 | 1R190 |
| 93 | 0R94 | 1R30 | 190 | 0R157 | 1R192 |
| 94 | 0R71 | 0R115 | 191 | 0R41 | 1R148 |
| 95 | 0R116 | 1R94 | 192 | 0R128 | 1R193 |
| 96 | 0R50 | 1R105 | 193 | 0R143 | 1R148 |

## C.2 PA194Q6

| State | Read 0 | Read 1 | State | Read 0 | Read 1 |
|---|---|---|---|---|---|
| 0 | 1R1 | 1R2 | 97 | 0R117 | 1R97 |
| 1 | 1R3 | 1R1 | 98 | 0R30 | 1R49 |
| 2 | 0R0 | 1R2 | 99 | 0R48 | 1R61 |
| 3 | 0R4 | 0R1 | 100 | 0R87 | 1L118 |
| 4 | 1R5 | 1L6 | 101 | 0R106 | 0L14 |
| 5 | 1L7 | 1L8 | 102 | 0R36 | 1R119 |
| 6 | 0L7 | 1L6 | 103 | 0R118 | 1R79 |
| 7 | 0L9 | 1L6 | 104 | 0L46 | 0R120 |
| 8 | 0L10 | 0L11 | 105 | 0L45 | 1R121 |
| 9 | 1R12 | 0L9 | 106 | 0L96 | 1R122 |
| 10 | 0L9 | 0L10 | 107 | 0R123 | 0L14 |
| 11 | 1L8 | 1L11 | 108 | 0R72 | 1R108 |
| 12 | 0R12 | 0R13 | 109 | 0R120 | 1R109 |
| 13 | 0L14 | 0R13 | 110 | 0L46 | 0R108 |
| 14 | 1L15 | 0L15 | 111 | 0R71 | 0R97 |
| 15 | 0L15 | 1L16 | 112 | 0R59 | 0R97 |
| 16 | 0L17 | 1L16 | 113 | 0L65 | 1R124 |
| 17 | 0L18 | 1L16 | 114 | 0R125 | 1R126 |
| 18 | 0L19 | 1L16 | 115 | 0R109 | 1R115 |
| 19 | 0L20 | 1L16 | 116 | 0R127 | 1R128 |
| 20 | 0L21 | 1L16 | 117 | 0R115 | 1R117 |
| 21 | 0L22 | 1L16 | 118 | 0R129 | 1L67 |
| 22 | 0L23 | 1L16 | 119 | 0R130 | 1R131 |
| 23 | 0R23 | 1R24 | 120 | 0R14 | 1R120 |
| 24 | 0R25 | 1R26 | 121 | 0R94 | 0L14 |
| 25 | 0R27 | 1R28 | 122 | 0R96 | 1L100 |
| 26 | 0R29 | 1R30 | 123 | 0R132 | 1R132 |
| 27 | 0R31 | 1RH | 124 | 0R133 | 1R134 |
| 28 | 0L32 | 1R33 | 125 | 0R32 | 1R36 |
| 29 | 0R34 | 1R35 | 126 | 0R31 | 1R80 |
| 30 | 0L36 | 1L37 | 127 | 0R135 | 1R136 |
| 31 | 0R38 | 1R39 | 128 | 0R137 | 1R138 |
| 32 | 0R40 | 1R41 | 129 | 0R59 | 0R117 |
| 33 | 0R42 | 1R43 | 130 | 0R104 | 1R121 |
| 34 | 0R44 | 1R45 | 131 | 0R83 | 1R82 |
| 35 | 0R46 | 1R47 | 132 | 0R94 | 1R48 |
| 36 | 0R48 | 1R49 | 133 | 0R138 | 1R139 |
| 37 | 0R50 | 1R51 | 134 | 0R140 | 1R141 |
| 38 | 0R52 | 1R42 | 135 | 0R102 | 1R142 |
| 39 | 0R53 | 1R54 | 136 | 0L138 | 1R94 |
| 40 | 0R55 | 1R56 | 137 | 0R41 | 1L143 |
| 41 | 0R57 | 1R58 | 138 | 0R144 | 1R145 |
| 42 | 0R59 | 0R60 | 139 | 0R146 | 1R147 |
| 43 | 0R61 | 0L14 | 140 | 0R148 | 1R91 |
| 44 | 0R62 | 1R63 | 141 | 0R149 | 0L10 |
| 45 | 0R55 | 1R64 | 142 | 0R91 | 1R64 |
| 46 | 0L65 | 1R66 | 143 | 0R148 | 1R145 |
| 47 | 0R36 | 1R67 | 144 | 0R102 | 1R67 |
| 48 | 0L68 | 1L69 | 145 | 0R150 | 1R48 |
| 49 | 0R70 | 0L14 | 146 | 0R151 | 1R152 |
| 50 | 0R71 | 0R72 | 147 | 0R153 | 1R154 |

| State | Read 0 | Read 1 | State | Read 0 | Read 1 |
|---|---|---|---|---|---|
| 51 | 0R73 | 0L49 | 148 | 0R136 | 1R41 |
| 52 | 0R74 | 1R54 | 149 | 0R155 | 1R156 |
| 53 | 0L75 | 0L15 | 150 | 0L157 | 1L153 |
| 54 | 0R76 | 0L10 | 151 | 0R28 | 1R158 |
| 55 | 0R77 | 1R78 | 152 | 0R145 | 1R159 |
| 56 | 0R79 | 1R80 | 153 | 0R137 | 1R160 |
| 57 | 0R81 | 1R82 | 154 | 0R161 | 1R162 |
| 58 | 0R83 | 1R84 | 155 | 0R31 | 1R53 |
| 59 | 0L85 | 0R1 | 156 | 0R163 | 1R155 |
| 60 | 0R86 | 1R60 | 157 | 0R164 | 1R165 |
| 61 | 0L77 | 1L68 | 158 | 0R166 | 1R167 |
| 62 | 0R87 | 0L7 | 159 | 0L167 | 1R168 |
| 63 | 0R88 | 1R89 | 160 | 0R169 | 1R170 |
| 64 | 0R42 | 1R90 | 161 | 0R171 | 1R171 |
| 65 | 0R91 | 1R49 | 162 | 0R172 | 1R173 |
| 66 | 0R92 | 0L76 | 163 | 0R98 | 1R118 |
| 67 | 0R68 | 1R33 | 164 | 0R80 | 1R172 |
| 68 | 0R74 | 1R93 | 165 | 0R135 | 1R172 |
| 69 | 0R94 | 1R95 | 166 | 0R174 | 1R175 |
| 70 | 0L96 | 1L33 | 167 | 0R176 | 1R165 |
| 71 | 0L85 | 0R2 | 168 | 0R145 | 1R138 |
| 72 | 0R97 | 1R72 | 169 | 0R177 | 1R178 |
| 73 | 0R76 | 1R98 | 170 | 0R179 | 1R178 |
| 74 | 0R59 | 0R86 | 171 | 0L116 | 1R127 |
| 75 | 0R68 | 1R99 | 172 | 0L169 | 1L180 |
| 76 | 0R42 | 0L13 | 173 | 0R143 | 1R180 |
| 77 | 0R87 | 1R37 | 174 | 0R144 | 1R181 |
| 78 | 0R30 | 1R100 | 175 | 0R159 | 1R170 |
| 79 | 0R42 | 1R101 | 176 | 0R182 | 1R183 |
| 80 | 0R102 | 1R103 | 177 | 0R135 | 1R150 |
| 81 | 0R104 | 1L105 | 178 | 0R144 | 1R136 |
| 82 | 0R30 | 1R81 | 179 | 0R184 | 1R41 |
| 83 | 0R99 | 1R106 | 180 | 0R185 | 1R179 |
| 84 | 0R30 | 1R107 | 181 | 0R186 | 1R172 |
| 85 | 1R85 | 1L15 | 182 | 0R144 | 1R184 |
| 86 | 0R108 | 1R86 | 183 | 0R187 | 1R187 |
| 87 | 0L46 | 0R109 | 184 | 0R135 | 1R137 |
| 88 | 0R110 | 1R100 | 185 | 0R171 | 1R188 |
| 89 | 0R110 | 1R70 | 186 | 0R150 | 1R189 |
| 90 | 0R111 | 0L14 | 187 | 0R69 | 1L190 |
| 91 | 0R112 | 1L75 | 188 | 0R191 | 1R41 |
| 92 | 0R113 | 1R114 | 189 | 0R32 | 1R190 |
| 93 | 0R94 | 1R30 | 190 | 0R157 | 1R192 |
| 94 | 0R71 | 0R115 | 191 | 0R41 | 1R148 |
| 95 | 0R116 | 1R94 | 192 | 0R128 | 1R193 |
| 96 | 0R50 | 1R105 | 193 | 0R143 | 1R148 |